



Product Dissection for Swiggy

Company Overview:

Swiggy, founded in 2014 by Sriharsha Majety, Nandan Reddy, and Rahul Jaimini, has successfully pioneered the hyperlocal commerce industry in India. The company launched its food delivery service in 2014 and expanded into quick commerce with Instamart in 2020.

Starting its operations in Bengaluru, Swiggy now serves millions of consumers by partnering with over 280,000 restaurants across more than 600 cities in India. Its reliable services, user-friendly platform, and fast delivery network have made Swiggy one of India's leading on-demand delivery platforms.

Product Dissection and Real-World Challenges Solved by Swiggy:

Swiggy has transformed the food delivery ecosystem addressing several real-world challenges through providing a seamless digital experience that connects customers, restaurants, and delivery partners efficiently.

One of the major problems solved by Swiggy is the difficulty in accessing restaurant food conveniently from home. Earlier, customers had limited food options and relied on phone-based ordering systems, which were often slow and inefficient. Swiggy simplified this process by offering a centralised platform where users can browse restaurants, explore menus, place orders, and track deliveries in real time.

Another challenge addressed by the platform is delayed and unreliable delivery. With its extensive delivery partner network and live order tracking system, the platform ensures faster and more transparent deliveries. Customers receive real time updates regarding order preparation, dispatching, expected delivery time which improves trust and user satisfaction.

The platform also provides different payment options like UPI, debit/credit cards, cash on delivery for customer convenience. It also analyses ratings, reviews, personalised recommendations provided by customers to improve decision-making and user experience.

In addition, Swiggy expanded into quick commerce with Instamart to cater the growing demand for instant grocery and daily essentials delivery. This innovation helps users to access essential products within minutes.

Thus, Swiggy's technology-driven ecosystem effectively solves challenges related to food accessibility, delivery efficiency, restaurant reach, and customer convenience, making it one of India's leading hyperlocal delivery platforms.

Case Study: Real-World Problems and Swiggy's Innovative Solutions

Swiggy, one of the leading delivery platforms in India, has not only revolutionised the food delivery industry but has also addressed several real-world challenges through its innovative and reliable services. By identifying the user needs and leveraging technology, Swiggy has evolved into a solution-driven platform that effectively tackles issues related to food ordering, delayed delivery, limited restaurant accessibility and customer convenience.

Problem 1: Difficulty in Ordering Food Conveniently

Real-World Challenge: In earlier times, customers often faced difficulty in ordering food from restaurants from home due to limited delivery options, long waiting times, and inefficient phone-based ordering systems.

Swiggy's Solution: Swiggy addresses this issue by introducing a centralised digital platform where users can browse nearby restaurants, explore menus, place order instantly, and make order payments conveniently. This solution by the platform has simplified the food ordering process and improved customer experience.

Problem 2: Delayed and Unreliable Deliveries

Real-World Challenge: Traditional food delivery systems lacked transparency and often resulted in delayed deliveries. This caused customer dissatisfaction and low ratings.

Swiggy's Solution: Swiggy uses a large delivery partner network with real-time order tracking system. The platform also optimises route management to ensure faster and more reliable deliveries. Customers can also track their order live from restaurant preparation to doorstep delivery. This ensures transparency and reliability in food delivery while significantly improving the overall user experience.

Problem 3: Limited Visibility for Restaurants

Real-World Challenge: Many small and local restaurants struggled to reach a large customer base. This may be due to limited marketing strategies and delivery infrastructure.

Swiggy's Solution: Swiggy provides platform to these restaurants to showcase menus, attract customers, manage online orders, and increase sales without building their own delivery system. The platform allows restaurants to focus more on food quality and marketing strategies rather than managing their own delivery infrastructure.

Problem 4: Inconvenient Payment Methods

Real-World Challenge: While ordering food, customers often experienced inconvenience due to limited and inconsistent payment options offered by different restaurants.

Swiggy's Solution: Swiggy has resolved this problem by integrating multiple payment options such as UPI, credit/debit cards, digital wallets, and cash on delivery on the platform. This makes the transactions smooth and flexible for customers, that improves overall customer satisfaction.

Problem 5: Urgent Need for Grocery Delivery

Real-World Challenge: People often require groceries and daily essentials urgently but may not have the time to visit the store physically. This created a growing need for a platform that allows users to conveniently order essential items online.

Swiggy's Solution: By launching Swiggy Instamart, Swiggy effectively addressed this challenge. Instamart enables users to order groceries and daily essentials online and

receive the delivery within minutes without visiting physical stores. This service improves customer convenience and saves valuable time.

Conclusion:

Swiggy's journey from a food delivery application to one of India's leading hyperlocal commerce platforms is a testament to its ability to identify real-world challenges and provide innovative solutions. By creating a platform where users can conveniently order food, explore restaurants and menus, make seamless payments, and track food preparation and delivery in real time, Swiggy has addressed various challenges that users encountered while ordering food.

This case study highlights how Swiggy's user-centric approach, technological innovation, and efficient delivery network have positioned it as a leader in the hyperlocal commerce industry. Through its reliable services, the platform has effectively transformed the way people order food and daily essentials online.

Top Features of Swiggy:

- 1. User Accounts and Profiles:** Swiggy allows users to create personal accounts using mobile numbers or email addresses. Users can also manage personal details, saved addresses, payment methods, and previous orders through profiles.
- 2. Restaurant Listings and Search:** Swiggy provides a wide range of restaurant options where users can explore restaurants based on filters such as location, cuisine, ratings, pricing, and reviews. These options improve food discovery and user convenience.
- 3. Live Order Tracking:** Swiggy offers real-time order tracking system that allows customers to monitor order preparation, dispatch, delivery partner information, and expected delivery time. This improves transparency and enhances customer trust.
- 4. Multiple Payment Options:** Swiggy offers wide range in selecting payment options such as UPI, credit/debit cards, digital wallets, and cash on delivery, ensuring flexibility and convenience for users.
- 5. Ratings and Reviews:** Users can provide ratings and reviews for restaurants and delivery experiences. This helps maintain service quality and also assists other users to make informed decisions regarding restaurants.

6. **Offers and Discounts:** Swiggy provides promotional offers, coupon codes, cashback rewards, and discounts to attract customers and user engagement.
7. **Swiggy Instamart:** With Swiggy Instamart, users can order groceries and daily essentials online and receive deliveries within minutes, expanding Swiggy's services beyond food delivery.

Schema Description:

The schema design of Swiggy consists of multiple entities that represent the core components of the platform. These entities include Users, Restaurants, Menu Items, Orders, Order Items, Delivery Partners, Deliveries, Payments, and Reviews. Each entity has specific attributes that describes its properties and relationships with other entities. With the help of these entities Swiggy efficiently manage its core operations such as ordering food, restaurants visibility, tracking deliveries, payment options, and customer feedback.

User Entity:

Users are the primary customers of the platform who browse restaurants, place orders, make payments, and provide reviews. The User entity contains the information of the user.

- **User ID (Primary Key):** Unique Identifier for each user.
- **Name:** Full name of the user.
- **Email:** User's email address.
- **Phone Number:** User's registered contact number.
- **Address:** Delivery address provided by the user.
- **Registration Date:** Date when the user created the account.

Restaurant Entity:

The Restaurant entity contains the information about the food vendors who are registered on the platform.

- **Restaurant ID:** Unique identifier for each restaurant.
- **Restaurant Name:** Contains the full name of the restaurant.
- **Location:** Restaurant's address/location.
- **Cuisine Type:** Cuisine offered by restaurant.
- **Rating:** Average rating of the restaurant provided by customers.

- **Contact Number:** Contact details of the restaurant.

Menu Item Entity:

The Menu Item entity represents the food products offered by different restaurants.

- **Item ID:** Unique identifier for each food item.
- **Restaurant ID (Foreign Key referencing Restaurant entity):** Refers to restaurant offering that food item.
- **Item Name:** Indicates full name of the food item.
- **Category:** Veg, Non-Veg, Beverage, Dessert, etc.
- **Price:** Refers to the pricing of food item.
- **Availability Status:** Indicates whether the food item is available or not.

Order Entity:

The Order entity stores the information about the food products that are ordered by the customers.

- **Order ID:** Unique identifier for each order.
- **User ID (Foreign Key referencing User entity):** Refers to user who placed the order.
- **Restaurant ID (Foreign Key referencing Restaurant entity):** Refers to the restaurant the food item is ordered from.
- **Order Date:** Indicates the date and time of order placement.
- **Total Amount:** Refers to the total bill amount of the placed order.
- **Order Status:** Placed, Preparing, Out for Delivery, Delivered, Cancelled.

Order Item Entity:

The Order Item entity refers to the information regarding the items placed in the order by the customer.

- **Order Item ID (Primary Key):** Unique identifier assigned to each item placed in the order.
- **Order ID (Foreign Key referencing Order entity):** Refers to the placed order.
- **Item ID (Foreign Key referencing Menu Item entity):** Refers to the ordered item.
- **Quantity:** Indicates total number of units ordered by the customer.
- **Item Total:** Represents total cost of the items.

Delivery Partner Entity:

The Delivery Partner entity stores the information of the delivery partners who are responsible for delivering orders to the customers.

- **Delivery Partner ID (Primary Key):** Unique identifier for each delivery partner.
- **Name:** Full name of delivery partner.
- **Phone Number:** Contact details of the delivery partner.
- **Vehicle Type:** Bike, Scooter, etc.
- **Availability Status:** Refers whether the delivery partner is busy or available.

Delivery Entity:

The Delivery entity stores the delivery-related information such as delivery status, delivery time, etc. to provide transparency in delivery.

- **Delivery ID (Primary Key):** Unique identifier for each delivery.
- **Order ID (Foreign Key referencing Order entity):** Refers to the placed order.
- **Delivery Partner ID (Foreign Key referencing Delivery Partner entity):** Refers to the delivery partner who is delivering the order.
- **Delivery Status:** Assigned, Picked Up, Delivered.
- **Pickup Time:** Indicates the time order picked up by delivery partner.
- **Delivery Time:** Indicates the time order delivered to the customer.

Payment Entity:

The Payment entity registers the information related to customer transactions associated with orders.

- **Payment ID (Primary Key):** Unique identifier for each payment made by customer.
- **Order ID (Foreign Key referencing Order entity):** Refers to the placed order.
- **Payment Method:** UPI, debit/credit cards, digital wallets, or cash on delivery.
- **Payment Status:** Successful, Pending, Failed.
- **Payment Date:** Stores the date of payment.

Review Entity:

The Review entity allow customers to share feedback about restaurants and services which helps other customers to make informed decisions.

- **Review ID (Primary Key):** Unique identifier for each review or feedback.
- **User ID (Foreign Key referencing User entity):** Refers to the user who shared the feedback.

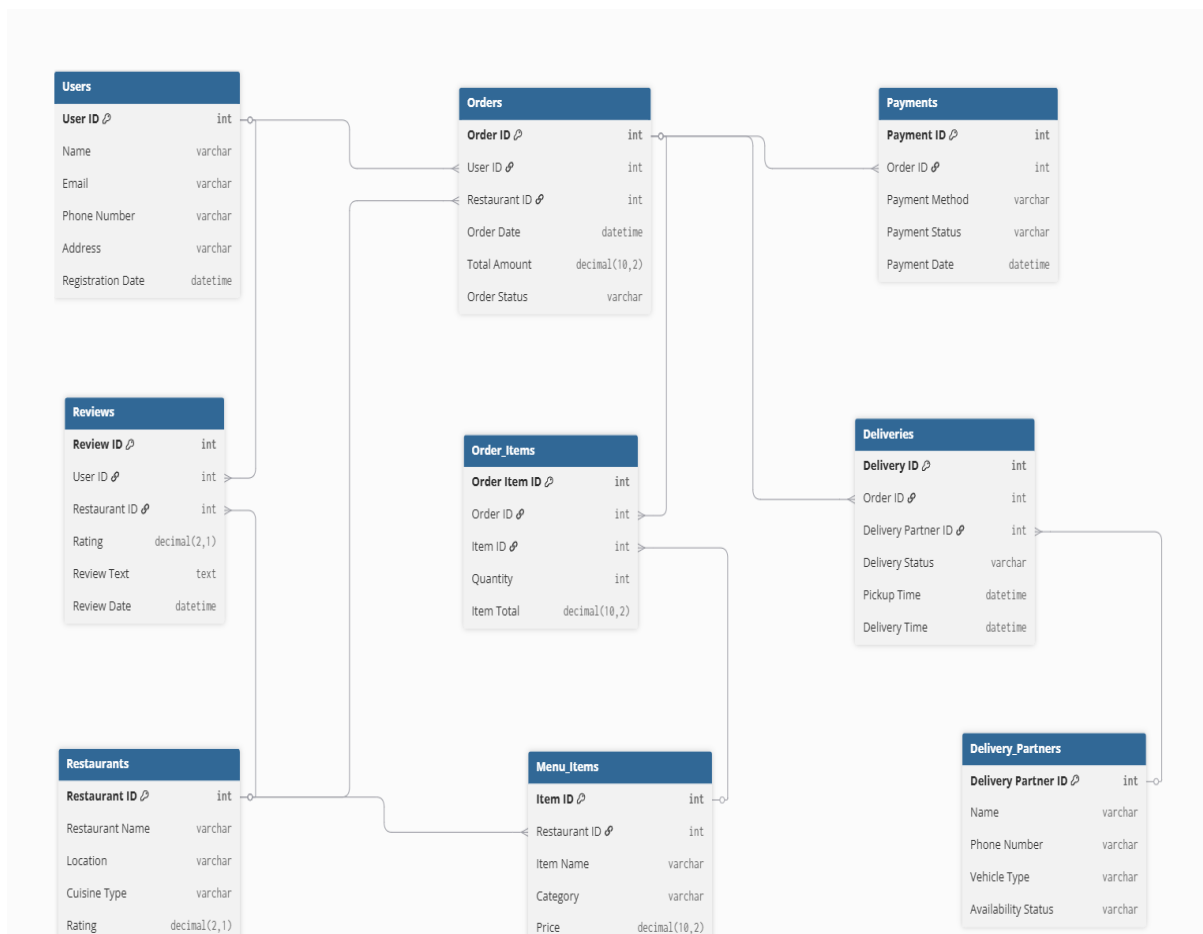
- **Restaurant ID (Foreign Key referencing Restaurant entity):** Refers to the restaurant regarding which the feedback is shared.
- **Rating:** Indicates the rating given by customer.
- **Review Text:** Written feedback provided by customer.
- **Review Date:** Indicates the date of review submission.

Relationships Between Entities:

- **Users place Orders:** Each user can place multiple orders.
- **Restaurants offer Menu Items:** Each restaurant offers multiple menu items which later ordered by users.
- **Order contains multiple Order Items:** Each order can contain multiple menu items from different restaurants.
- **Users make Payments for Orders:** User can use multiple payment options for order payments.
- **Delivery Partners deliver Orders:** Each delivery partner delivers order from different restaurants.
- **Users write Reviews for Restaurants:** Every user can share multiple reviews for restaurants about their service quality.

ER Diagram:

Now, let's construct the **ER(Entity-Relationship) diagram** that vividly portrays the relationships and attributes of the entities present within the Swiggy schema. This ER diagram serves as a visual representation of the platform's data model, highlighting the key entities, attributes, and relationships that support Swiggy's operations. By analysing this diagram, we gain a clearer understanding of the intricate interactions and connections between different entities that drive the platform's functionality and overall user experience.



Entity–Relationship (ER) Diagram of the Swiggy Database Schema

Conclusion:

In this case study, we explored the design of Swiggy’s database schema and Entity-Relationship (ER) diagram. Swiggy has revolutionised the hyperlocal commerce industry by transforming the way people order food and daily essentials online. The platform’s well-structured data model, consisting of entities such as Users, Restaurants, Menu Items, Orders, Order Items, Delivery Partners, Deliveries, Payments, and Reviews, forms the foundation for its seamless operations and user experience.

By analysing the schema and relationship between these entities, we gain valuable insights into how Swiggy effectively manages its operations. The schema design ensures data consistency and smooth interaction among all stakeholders within the platform. of user while ordering food online and delivering food, contributing to its widespread popularity and continued growth in the delivery industry. Overall, this case study demonstrates Swiggy’s effective operational efficiency, widespread popularity, and continued growth in hyperlocal commerce industry.